Vorlesung 13

Magnetische Dipolnäherung (fortf.)

$$\boxed{\int \vec{J}(\vec{r}\,')\,\vec{\nabla}' f(\vec{r}\,')\,d^3r' = 0} \quad (*)$$

Vorl. 12

$f(\vec{r}\,') = x'y'$: $\quad \vec{\nabla}' f = \begin{pmatrix} y' \\ x' \\ 0 \end{pmatrix} \quad \overset{\text{Gl.}(*)}{\Rightarrow} \quad \int (J_x y' + J_y x')\,d^3r' = 0$

$f(\vec{r}\,') = y'z'$: $\quad \dots \quad \Rightarrow \quad \int (J_y z' + J_z y')\,d^3r' = 0$

$f(\vec{r}\,') = x'z'$: $\quad \dots \quad \Rightarrow \quad \int (J_z x' + J_x z')\,d^3r' = 0$

A

$f(\vec{r}) = \frac{1}{2} x'^2$: $\quad \vec{\nabla}' f = \begin{pmatrix} x' \\ 0 \\ 0 \end{pmatrix} \quad \overset{(*)}{\Rightarrow} \quad \int J_x x'\,d^3r' = 0$

$\dots \quad \int J_y y'\,d^3r' = 0$

$\int J_z z'\,d^3r' = 0$

B

Vorl. 12

$$\vec{A} \approx \frac{\mu_0}{4\pi}\left[\frac{1}{r}\int_V \vec{J}(\vec{r}\,')\,d^3r' + \int_V d^3r'\,\vec{J}(\vec{r}\,')\,\frac{\vec{r}\cdot\vec{r}\,'}{r^3}\right] =$$

$$= \frac{\mu_0}{4\pi}\left[\frac{1}{r}\underbrace{\begin{pmatrix}\int J_x(\vec{r}\,')\,d^3r' \\ \int J_y(\vec{r}\,')\,d^3r' \\ \int J_z(\vec{r}\,')\,d^3r'\end{pmatrix}}_{\vec{I}_0} + \frac{1}{r^3}\underbrace{\begin{pmatrix}\int d^3r'\,J_x(\vec{r}\,')\,(\vec{r}\cdot\vec{r}\,') \\ \int d^3r'\,J_y(\vec{r}\,')\,(\vec{r}\cdot\vec{r}\,') \\ \int d^3r'\,J_z(\vec{r}\,')\,(\vec{r}\cdot\vec{r}\,')\end{pmatrix}}_{\vec{I}}\right]$$

$$I_\alpha = \vec{r} \cdot \int J_\alpha \, \vec{r}\,' \, d^3r' = \overset{\text{Skalarprodukt}}{\sum_\beta} r_\beta \int J_\alpha \, r'_\beta \, d^3r' = \overbrace{\underbrace{r_\alpha \int J_\alpha \, r'_\alpha}_{\text{keine Einst. Summe}} \, d^3r'}^{\overset{!}{=} 0 \;\Leftarrow\; (B)} + \sum_{\substack{\beta \\ \beta \neq \alpha}} r_\beta \int J_\alpha \, r'_\beta \, d^3r =$$

$$= \sum_{\substack{\beta \\ \beta \neq \alpha}} r_\beta \int \left[\tfrac{1}{2} J_\alpha r'_\beta + \overbrace{\tfrac{1}{2} J_\alpha r'_\beta + \tfrac{1}{2} J_\beta r'_\alpha}^{\overset{!}{=} 0 \;\Leftarrow\; (A)} - \tfrac{1}{2} J_\beta r'_\alpha \right] d^3r'$$

$$\boxed{I_\alpha = \frac{1}{2} \sum_{\substack{\beta \\ \beta \neq \alpha}} r_\beta \int \left(J_\alpha r'_\beta - J_\beta r'_\alpha \right) d^3r'}$$

$$\vec{c} = \vec{J} \times \vec{r}\,' = \begin{pmatrix} J_y z' - J_z y' \\ J_z x' - J_x z' \\ J_x y' - J_y x' \end{pmatrix}$$

$$\vec{r} \times \vec{c} = \begin{pmatrix} y c_z - z c_y \\ z c_x - x c_z \\ x c_y - y c_x \end{pmatrix}$$

$$\begin{pmatrix} y \overset{c_z}{(J_x y' - J_y x')} + z \overset{-c_y}{(J_x z' - J_z x')} \\ x \overset{-c_z}{(J_y x' - J_x y')} + z \overset{c_x}{(J_y z' - J_z y')} \\ x \overset{c_y}{(J_z x' - J_x z')} + y \overset{-c_x}{(J_z y' - J_y z')} \end{pmatrix}$$

Letztendlich

$$\boxed{\vec{A}(\vec{r})} \simeq \frac{\mu_0}{4\pi}\,\frac{1}{r^3}\,\frac{1}{2}\,\vec{r}\times\int(\vec{J}\times\vec{r}\,')\,d^3r' = \boxed{\frac{\mu_0}{4\pi}\,\frac{1}{r^3}\left[\frac{1}{2}\int(\vec{r}\,'\times\vec{J})\,d^3r'\right]\times\vec{r}}$$

$$\boxed{\vec{m} \equiv \frac{1}{2}\int(\vec{r}\,'\times\vec{J})\,d^3r'}$$ $\leftarrow$ das magnetische Dipolmoment

$$\boxed{\vec{A}(\vec{r}) \simeq \frac{\mu_0}{4\pi}\,\frac{\vec{m}\times\vec{r}}{|\vec{r}|^3}}$$ $\leftarrow$ Magnetsche Dipolnäherung

$$\boxed{\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\frac{\vec{m}\times\vec{r}}{|\vec{r}|^3}}$$

$$\Downarrow$$

$$\vec{B}(\vec{r}) = \vec{\nabla}\times\vec{A} = \frac{\mu_0}{4\pi}\,\vec{\nabla}\times(\vec{m}\times\underset{\frac{\vec{r}}{r^3}}{\underbrace{\vec{F}}}) = \frac{\mu_0}{4\pi}\Big[\underbrace{\overbrace{(\vec{\nabla}\cdot\vec{F})}^{=0}}\,\vec{m} - \underbrace{(\vec{\nabla}\cdot\vec{m})\,\vec{F}}\Big]$$

konstante

$$\underbrace{\vec{a}\times(\vec{b}\times\vec{c})}_{=\,(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}}$$

$\vec{\nabla}$ muss links stehen

$$\vec{\nabla}\cdot\frac{\vec{r}}{r^3} = 4\pi\,\delta(\vec{r})\Big|_{\vec{r}\neq\vec{0}} = 0$$

(HA 1)

$$(\vec{\nabla}\cdot\vec{m})\,\vec{F} = (m_x\partial_x + m_y\partial_y + m_z\partial_z)\begin{pmatrix}\frac{x}{r^3}\\ \frac{y}{r^3}\\ \frac{z}{r^3}\end{pmatrix}\Bigg| \qquad r = \sqrt{x^2+y^2+z^2}$$

$$\partial_x \begin{pmatrix} \frac{x}{r^3} \\ \frac{y}{r^3} \\ \frac{z}{r^3} \end{pmatrix} = \begin{pmatrix} \frac{1}{r^3} + x\left(-3\frac{x}{r^5}\right) \\ y\left(-3\frac{x}{r^5}\right) \\ z\left(-3\frac{x}{r^5}\right) \end{pmatrix} = \frac{1}{r^3}\hat{n}_x - \frac{3x\vec{r}}{r^5} \quad \Big| \cdot m_x$$

$$\partial_y \vec{r} = \frac{1}{r^3}\hat{n}_y - \frac{3y\vec{r}}{r^3} \quad \Big| \cdot m_y$$

$$\partial_z \vec{r} = \frac{1}{r^3}\hat{n}_z - \frac{3z\vec{r}}{r^3} \quad \Big| \cdot m_z$$

$$\boxed{\vec{B}(\vec{r})} = -\frac{\mu_0}{4\pi}(\vec{m}\cdot\vec{\nabla})\vec{r} = \boxed{\frac{\mu_0}{4\pi}\left[3\frac{(\vec{m}\cdot\vec{r})\vec{r}}{r^5} - \frac{\vec{m}}{r^3}\right]}$$

$\uparrow$

nur Fernfeld $\leftarrow$ Isotroper Mittelwert $= 0$

Dipolnäherung : Nahfeld

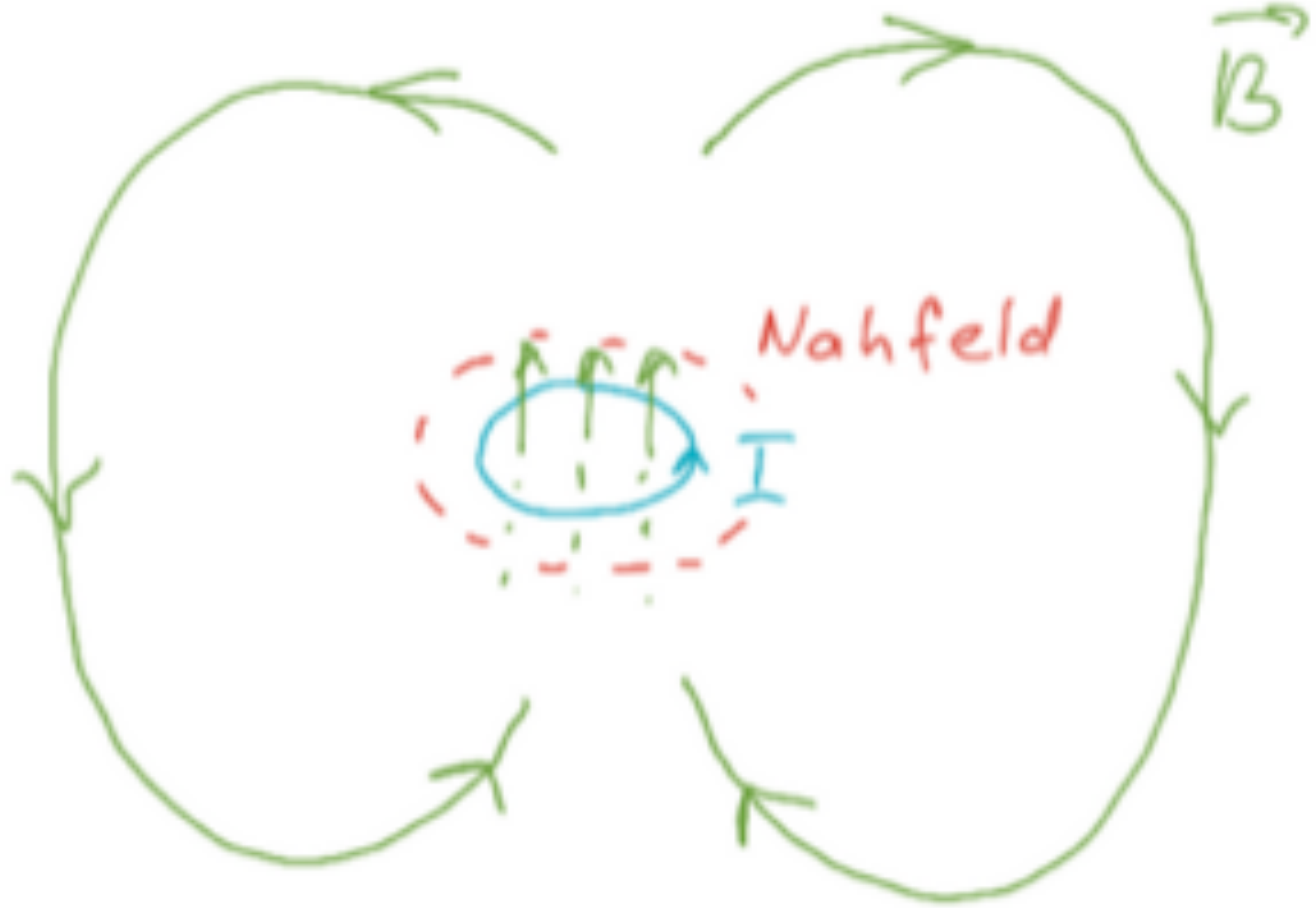
Bio-S. Gesetz

$$\int_V \vec{B}(\vec{r})\, d^3r = \int d^3r \,\frac{\mu_0}{4\pi}\int d^3r'\, \vec{J}(\vec{r}\,') \times \frac{\vec{r}-\vec{r}\,'}{|\vec{r}-\vec{r}\,'|^3} =$$

Kugel, R

$$= \frac{\mu_0}{4\pi}\int d^3r'\, \vec{J}(\vec{r}\,') \times \underbrace{\int d^3r \,\frac{\vec{r}-\vec{r}\,'}{|\vec{r}-\vec{r}\,'|^3}} = -\frac{\mu_0}{3}\, 2 \underbrace{\frac{1}{2}\int d^3r'\, \vec{J}(\vec{r}\,') \times \vec{r}\,'}_{-\vec{m}}$$

Vorl. 10: $\vec{I}_1 = -\frac{4\pi}{3}\vec{r}\,'$



$$\boxed{\int_V \vec{B}(\vec{r})\, d^3r = \frac{2}{3}\mu_0 \vec{m}}$$

→ Vollst. Formel für die magn. Dipolnäherung:

$$\boxed{\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi}\left[3\,\frac{(\vec{m}\cdot\vec{r})\vec{r}}{r^5} - \frac{\vec{m}}{r^3} + \underbrace{\frac{8\pi}{3}\vec{m}\,\delta(\vec{r})}\right]}$$

Plus

Fermi-Kontakt
Hyperfein
Wechselwirkung

Jackson 5.7
(zum Selbstlesen)

$\vec{H} = \frac{1}{\mu_0}\vec{B} - \vec{M}$

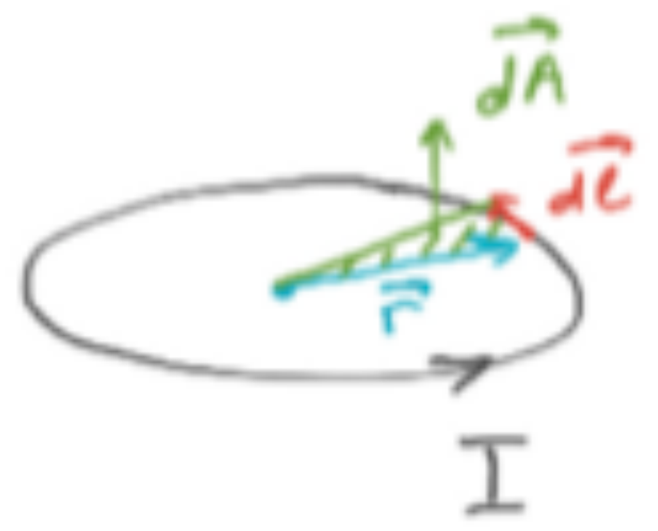
$\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$

$\Leftarrow$ $\vec{m}\uparrow$ $\vec{B}$ I

$\vec{p}\uparrow$ ⊕ ⊖ $\vec{E}$

$\left[\text{El. Dipol: } -\frac{4\pi}{3}\vec{p}\,\delta(\vec{r})\right]$

Magn. Moment für eine flache Stromschleife

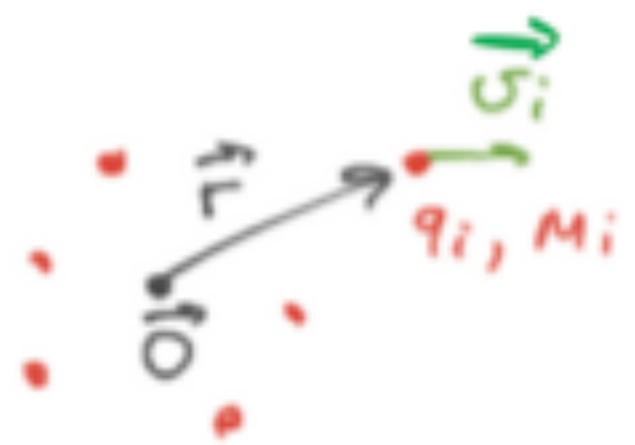


$$\vec{J}\, d^3r = \underbrace{\vec{J}\, da}_{\int = I}\, d\ell = I\, \vec{d\ell}$$

$$\boxed{\vec{m} = \frac{1}{2} \int \vec{r} \times \vec{J}(\vec{r})\, d^3\vec{r} = I \int \underbrace{\frac{1}{2}\, \vec{r} \times \vec{d\ell}}_{d\vec{A}} = I \vec{A}}$$

$$|d\vec{A}| = \underbrace{\frac{1}{2}\, r \cdot d\ell \sin\alpha}_{\text{Fläche des Dreiecks}}$$

Magn. Moment für ein System der Punktteilchen



$$\vec{J} = \rho(\vec{r})\, \vec{v}(\vec{r}) \rightarrow \sum_i q_i\, \delta(\vec{r} - \vec{r}_i)\, \vec{v}_i$$

$$\vec{m} = \frac{1}{2} \sum_i q_i\, \underbrace{\vec{r}_i \times \vec{v}_i\, \frac{M_i}{M_i}}_{\text{Drehimpuls } \vec{L}_i} = \sum_i \underbrace{\frac{q_i}{2 M_i}}_{\text{gyromagnetisches Verhältnis}} \vec{L}_i$$

Zum Weiterlesen:

Jackson: 5.2-5.4, 5.6, 5.7